TANMOY ROY CHAMAK

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PROFILE

With a background in Computer Science and Engineering, I have built a strong foundation in software development and intelligent systems. I am skilled in programming languages such as C, C++, Python, and Java, and have hands-on experience with IoT, Linux, and object-oriented programming. I am particularly interested in applying and expanding my knowledge in areas related to Artificial Intelligence and Machine Learning. Whether through research or practical development, I am driven to contribute to innovative solutions and continuously enhance my expertise in emerging technologies.

EXPERIENE

SAMSUNG Research & Development Institute — Bangalore, Karnataka, India

Research Intern — Dec 2022 – Aug 2023

Project: SAMSUNG PRISM [TizenRT | Binary Compression Optimization] — **Technologies:** C, TizenRT, Linux, IoT

Completed this internship (Remote) under the Samsung PRISM Program, where I worked on enhancing the TizenRT operating system through the implementation of new binary compression techniques.

Responsibilities:

- Analyzed TizenRT OS architecture and existing compression techniques.
- Researched binary compression methods through academic literature.
- Integrated a new compression method for the TizenRT operating system.
- Evaluated performance gains on a TizenRT development board across multiple applications.

EDUCATION

B.Tech in Computer Science and Engineering [CGPA 8.83 / 10] — 2020 – 2024

Vellore Institute of Technology, Vellore — Tamil Nadu, India

B.Sc in Computer Science and Engineering [Transferred to VIT] — 2018 – 2020

Rajshahi University of Engineering and Technology — Rajshahi, Bangladesh

Higher Secondary School Certificate [GPA 5.0 / 5.0] — 2016 – 2018

Rajshahi College — Rajshahi, Bangladesh

Secondary School Certificate [GPA 5.0 / 5.0] — 2014 – 2016

Rajshahi Collegiate School — Rajshahi, Bangladesh

PUBLICATIONS

Multi-View Ensemble of Fine-Tuned CNNs for Alzheimer's Disease Detection from 3D MRI — Under Review

Array (Elsevier)

Tanmoy Roy Chamak, Malay Yatin Sahasrabudhe

Developed a deep-learning approach for early Alzheimer's disease detection using 3D MRI data and transfer learning. Implemented a multi-plane slice extraction method and generated subject-level predictions through an ensemble model combining axial, coronal, and sagittal views. Used a strict subject-level dataset split to prevent data leakage. Achieved an overall accuracy of 83.10%, outperforming previous subject-level studies on the OASIS dataset. This multi-view ensemble technique enhanced both the accuracy and reliability of early AD detection.

TizenRT OS Based Data Compression Algorithms for IoT Devices — 2023

2023 International Conference on Computing, Communication, and Intelligent Systems (ICCCIS)

DOI: 10.1109/ICCCIS60361.2023.10425386

Tanmoy Roy Chamak, Sachi Kaushik, Shiven Singhal, Aditya Dadheech, Prof. A. Anitha, Prof. S. Jenicka, Vidisha Thapa

In the IoT world, managing data efficiently is crucial. TizenRT is an open-source system tailored for resource-limited IoT devices. This paper explores the integration of LZ4, a high-performance compression algorithm, into the TizenRT OS, alongside MiniZ and LZMA. The study conducted an in-depth comparative analysis to explore the performance, resource utilization, and adaptability of these algorithms within the TizenRT ecosystem, offering insights into improving TizenRT for IoT data management.

PROJECTS

Personal Portfolio Website | NextJS, TypeScript, Tailwind CSS, Vercel Jun 2025 – Nov 2025

My personal portfolio website is developed using NextJS, TypeScript, and Tailwind CSS, and deployed on Vercel. It features a modular, highly customizable architecture with a minimalistic and user-friendly design. Built as a multi-page website, it effectively organizes content for both professional and academic purposes, ensuring clarity, simplicity, and a seamless browsing experience.

Little Lemon Cross Platform Mobile App | Expo, React Native, JavaScript Aug 2025 – Sep 2025

Completed for the **"Meta React Native Specialization"**, this capstone involved building a cross-platform mobile app using React Native. Implemented modular components, API consumption, responsive layouts, and platform-adaptive UI. Demonstrated strong JavaScript mobile development and component-based architecture.

Little Lemon Android App | Kotlin, Material UI, Jetpack Compose Jul 2025 – Aug 2025

Built as part of the **"Meta Android Developer Professional Certificate"**, this capstone project focused on creating a functional Android application with modern UI components, structured navigation, lifecycle handling, and clean architecture. Displayed strong Android development and mobile UI/UX skills.

Little Lemon Database | MySQL, MySQL Workbench, Python Apr 2025 – May 2025

Created during the **"Meta Database Engineer Professional Certificate"**, this capstone project involved designing a complete relational database system with ER diagrams, advanced SQL queries, stored procedures, indexing, and performance tuning. Demonstrated mastery of database engineering and optimization.

Little Lemon Full Stack Back-End | Django, DRF, JavaScript, API Jan 2025 – Feb 2025

Developed as part of the **"Meta Back-End Developer Professional Certificate"**, this capstone project involved building a complete restaurant management system. Implemented RESTful APIs using Django REST Framework, token-based authentication, role-based permissions, and MySQL data persistence. Integrated a JavaScript front end to consume API endpoints, demonstrating strong back-end engineering and API development skills.

Little Lemon API | Django, DRF Jan 2025 – Jan 2025

Developed as part of the **"Meta Back-End Developer Professional Certificate"**, this project included building a production-ready REST API with menu management, booking endpoints, authentication, and role-based access control. Showcased strong skills in Django, database modeling, and secure API design.

Little Lemon Front-End | ReactJS, JavaScript, UI/UX Nov 2024 – Dec 2024

Completed for the **"Meta Front-End Developer Professional Certificate"**, this capstone focused on creating a responsive restaurant reservation web app. Built interactive booking components, implemented client-side validation, and integrated REST APIs for dynamic data flow. Demonstrated proficiency in React, state management, and modern UI/UX practices.

Alzheimer Detection Using Deep Learning | TensorFlow, CNN, Transfer Learning Jan 2024 – May 2024

The **"Undergraduate Capstone Project"** focused on developing a novel approach for early Alzheimer's disease detection, addressing data leakage through careful dataset preparation and subject-level splitting. The project applied various classification models to MRI slices from different orientations, evaluating performance at both slice and subject levels. By using ensemble learning, it combined predictions from multiple MRI orientations, improving accuracy to 83.10% and surpassing previous research on the OASIS dataset, offering a more accurate and reliable method for early Alzheimer's detection.

COVID-19 Data Visualization | MATLAB, Object Oriented Programming, GUI Sep 2020 – Sep 2020

Completed for the **"Mastering Programming with MATLAB"** Coursework, this project involved building an interactive application to visualize COVID-19 data. Used Object-Oriented Programming and MATLAB GUI components to process and display data from the Johns Hopkins dataset in real time.

CERTIFICATIONS

OCI Data Science Professional (Oracle Cloud Infrastructure)	Oct 2025
OCI Generative AI Professional (Oracle Cloud Infrastructure)	Oct 2025
OCI AI Foundations Associate (Oracle Cloud Infrastructure)	Oct 2025
Meta Data Analyst with GenAI Professional Certificate (Meta)	Oct 2025
Meta React Native Specialization (Meta)	Sep 2025
Meta Android Developer Professional Certificate (Meta)	Sep 2025
Meta Database Engineer Professional Certificate (Meta)	May 2025
Machine Learning Specialization (DeepLearning.AI, Stanford University)	Feb 2025
Meta Back-End Developer Professional Certificate (Meta)	Feb 2025
Meta Front-End Developer Professional Certificate (Meta)	Jan 2025
Google IT Support Professional Certificate (Google)	Nov 2024
MATLAB Programming for Engineers and Scientists Specialization (Vanderbilt University)	Oct 2020

TECHNICAL SKILLS

Programming Languages	: C, C++, Python, Java, MATLAB, JavaScript, PHP, R, HTML, CSS
Databases	: SQL*Plus, MySQL, MongoDB
IDEs and Version Control	: Visual Studio Code, IntelliJ IDEA, Scene Builder, Git, GitHub Desktop
Frameworks	: NodeJS, ReactJS, NextJS, Tailwind CSS, TensorFlow, Django, Django REST Framework (DRF), Flask
Software	: Adobe Photoshop, Autodesk AutoCAD, Microsoft Office Suite, Oracle VirtualBox, Microsoft PowerBI, Tableau, MySQL Workbench
Operating Systems	: Windows, Linux (Ubuntu)
Cloud & Deployment	: Google Cloud Platform, Amazon Web Service, Azure, Vercel,

ACHIEVEMENTS

Metaverse Ideathon Winner – Samsung PRISM — May 2024 – Jun 2024
Team **Wizardy Whizkids** won **Metaverse Ideathon** organized by **Samsung PRISM**

Certificate of Excellence – Samsung PRISM — Dec 2022 – Aug 2023
Awarded For **Outstanding Performance** during **Internship (Remote)** under **Samsung PRISM**

Merit Scholarship - Vellore Institute of Technology — Aug 2021 – Jul 2022
Awarded for the **Outstanding Performance** during the academic year 2021 – 2022

Merit Scholarship - Vellore Institute of Technology — Aug 2020 – Jul 2021
Awarded for the **Outstanding Performance** during the academic year 2020 – 2021

Study in India (SII) Scholarship — Aug 2020 – Jul 2024
Sponsored by **Ministry of Education (MoE), Government of India** to Pursue **B.Tech**

Ranked 10th in the Bangladesh Physics Olympiad 2018 Regional Round — Jan 2018 – Jan 2018
Ranked 10th in Category C (Class XI-XII) in the **Rajshahi Regional Round**

Education Board Scholarship — July 2016 – Jun 2018
Scholarship based on the result of **Secondary School Certificate Examination**

Education Board Scholarship — Jan 2014 – Dec 2015
Scholarship based on the result of **Junior School Certificate Examination**

Education Board Scholarship — Jan 2011 – Dec 2013
Scholarship based on the result of **Primary School Certificate Examination**

EXTRACURRICULAR ACTIVITIES

Cloud Bootcamp — Aug 2023 – Oct 2023
Program Authorized by **GeeksforGeeks** and Sponsored by **Google for Developers**

Mentorship Training - McMaster University — Jun 2021 – Jul 2021
Completed the mentor training with **McMaster Start Coding** and Mentored Three Children

Virtual Poster Exhibition - Vellore Institute of Technology — Jan 2021 – Jan 2021
Participated in the **Ethics and Values Poster Exhibition** on **Personal and Social Responsibility During the Pandemic**

LANGUAGE PROFICIENCY

English	: Professional Proficiency	**Hindi**	: Professional Proficiency
Bengali	: Native Proficiency	**Japanese**	: Elementary Proficiency

IMPORTANT LINKS

GitHub	: https://github.com/TANMOY-TRC	**HackerRank**	: https://www.hackerrank.com/profile/TANMOY_TRC
LeetCode	: https://leetcode.com/u/TANMOY-TRC	**Credly**	: https://www.credly.com/users/tanmoy-roy-chamak